Statistics

- ***Descriptive*:**
  - **Quantitative: 1) continues 2) discrete**

  - **Continues:**

    1. **continuous data is the data that can be of any value. Over time, some continuous data can change.**
    2. **It may take any numeric value, within a potential value range of finite or infinite.**
    3. **The continuous data can be broken down into fractions and decimals, i.e. according to measurement accuracy, it can be significantly subdivided into smaller sections.**
    4. **Continuous Data Examples: Measurement of height and weight of a student, Daily temperature measurement of a place, Wind speed measured daily, etc.**
    5. **measurable**

  - **Discrete:**

    1. **Data that can only take on certain values are discrete data.**
    2. **These values do not have to be complete numbers, but they are values that are fixed.**
    3. **It only contains finite values, the subdivision of which is not possible.**
    4. **It includes only those values which are separate and can only be counted in whole numbers or integers, which means that the data cannot be split into fractions or decimals.**

5. **Discrete Data Examples: The number of students in a class, the number of chocolates in a bag, the number of strings on the guitar, the number of fishes in the aquarium, etc.**
6. **Countable**
7. **To consider if we have continuous or discrete data, we should see if we can split our data into smaller and smaller units.**

- **To measure them we use 4 methods :**

  1. **Center**
  2. **Spread**
  3. **Shape**
  4. **Outliers**

- **We can divide center into 3 ways:**

  1. **Mean:**

     - **Is the result of dividing sum of set of numbers on their count**
     - **We can represent it by sigma shape or using x or y bar**

  2. **Median:**

     - **The middle number of group of sorted numbers**

  3. **Mode:**

     - **Is the most repeated number in a group of numbers**
     - **It can be set of numbers**

- **Categorical: 1) nominal 2) ordinal**
  - **Nominal:**
    1. **Nominal data simply names something without assigning it to an order in relation to other numbered objects or pieces of data.**
    2. **An example of nominal data might be a "pass" or "fail" classification for each student's test result.**
    3. **Nominal data provides some information about a group or set of events, even if that information is limited to mere counts.**
    4. **Nominal scales are also known as labels**
  - **Ordinal:**
    1. **Ordinal data, unlike nominal data, involves some order; ordinal numbers stand in relation to each other in a ranked fashion.**
    2. **For example, suppose you receive a survey from your favorite restaurant that asks you to provide**
    3. **feedback on the service you received. You can rank the quality of service as "1" for poor, "2" for below average, "3" for average, "4" for very good and "5" for excellent.**
    4. **The data collected by this survey are examples of ordinal data. Here the numbers assigned have an order or rank; that is, a ranking of "4" is better than a ranking of "2."**

Notation:

- **Common math language used to communicate regardless of spoken language. (Essential to communicating ideas regarding data)**
- **Variables:**
  - **Random: Notated by a capital letter (They have many different values)**
  - **Observed: Notated by a lowercase letter with a subscript (signify a specific value)**